

WASP-36b

R-0681

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_260208 · created 2026-07-27T19:35:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461079.7567 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.77 +/- 1.62 [%]
Orbital Inclination (inc)	87.39 +/- 2.73
Airmass coefficient 1 (a1)	1.157 +/- 0.049
Airmass coefficient 2 (a2)	-0.086 +/- 0.1
Scatter in the residuals of the lightcurve fit is	4.38 %
Best Comparison Star	#3 - [235, 204]
Optimal Aperture	3.29
Optimal Annulus	12.25
Transit Duration (day)	0.0927 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.166 $\pm$ 0.049 vs 0.1368 (deviation 0.45 σ)
Duration fit vs published	0.0927 d vs 0.0758 d (deviation 1.34 σ)
Mid-time O-C	0.7 min
Depth SNR	2.6
Residual scatter	4.38 %

Light curve

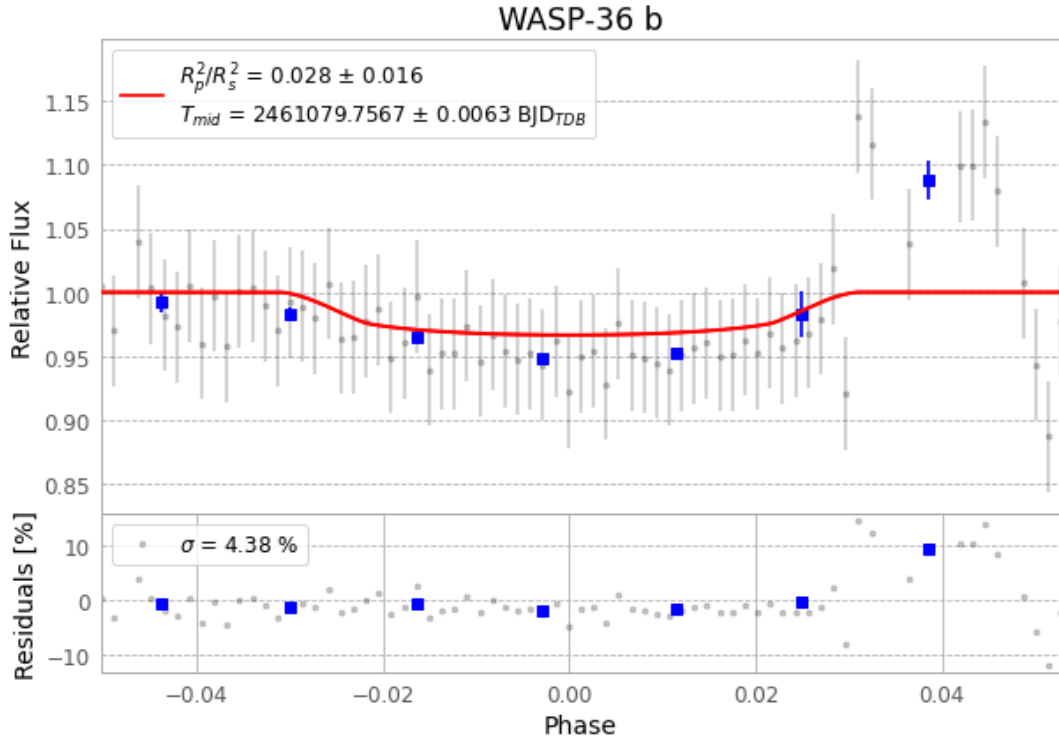


Figure 1 — FinalLightCurve_WASP-36 b_2026-02-07.png (SHA-256 f69b4a24660d27c4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff2283286e7cb14c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

77 FITS frames (51 MB), WASP-36260208040916.FITS → WASP-36260208075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260208.log (515e69a4f403...), FinalLightCurve_WASP-36 b_2026-02-07.png (f69b4a24660d...), FinalLightCurve_WASP-36 b_2026-02-07.csv (dfd11c65c9ae...)

Compute resources

Wall time 355.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:35Z.